

Research Notes on Policy gradient method

Policy gradient method

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi} [\nabla_{\theta} \log \pi_{\theta}(a | s) Q^{\pi}(s, a)]$$

>> Section 1

Methods like TRPO, PPO and natural policy gradient share a common idea - while the policy should be updated in the direction of the policy gradient, the update should be done in a safe and stable manner, typically measured by some distance with respect to the policy before the update. A similar notion of update stability is found in proximal convex optimization techniques like Mirror Descent. There, $\mathbf{x}$, the proposed minimizer of f in some constraint set C , is iteratively updated in the direction of the gradient ∇f , with a proximity penalty with respect to the current $\mathbf{x}_t$ measured by some Bregman divergence B_{ω} , which can be formalized by the following formula: $\mathbf{x}_{t+1} \in \arg \min_{\mathbf{x} \in C} \nabla f(\mathbf{x}_t)^T (\mathbf{x} - \mathbf{x}_t) + \eta B_{\omega}(\mathbf{x}, \mathbf{x}_t)$, where η controls the proximity between consecutive iterates, similar to the learning rate in gradient descent. This leads to reconsidering the policy update procedure as an optimization procedure aimed at finding an optimal policy, in the (non-convex) optimization landscape of the underlying Markov decision process (MDP). This optimization viewpoint of using the policy gradient is termed Mirror Descent Policy Optimization (MDPO), leading to the following update when the KL is the chosen Bregman divergence: $\pi_{t+1} \in \arg \max_{\pi} \mathbb{E}_{s, a \sim \pi_t} [\pi_t(s, a)] + \eta \mathbb{E}_{s, a \sim \pi_t} [D_{KL}(\pi || \pi_t)]$. With a parameterized policy π_{θ} , the MDPO loss becomes: $\max_{\theta} L(\theta, \theta_t) = \mathbb{E}_{s, a \sim \pi_{\theta_t}} [\pi_{\theta}(a | s) \pi_{\theta_t}(a | s) A \pi_{\theta_t}(s, a)] + \eta \mathbb{E}_{s, a \sim \pi_{\theta_t}} [D_{KL}(\pi_{\theta} || \pi_{\theta_t})]$. This objective can be used together with other common techniques like the clipping done in PPO. In fact, the KL divergence penalty also appears in the original PPO paper, suggesting the MDPO perspective as a theoretical unification of the main derivation concepts behind many concurrent policy gradient techniques.

>> Section 2

$\gamma \sum_{t \leq \tau \leq T} (\gamma \tau - t R_{\tau}) - b(S_t)$: used by the REINFORCE with baseline algorithm.

>> Section 3

Thus, we have an unbiased estimator of the policy gradient: $\nabla J(\theta) \approx \frac{1}{N} \sum_{n=1}^N \sum_{t=0}^T \nabla_{\theta} \ln \pi_{\theta}(A_t, n | S_t, n) (\gamma \tau - t R_{\tau})$ where the index n ranges over N rollout trajectories using the policy π_{θ} . The score function $\nabla_{\theta} \ln \pi_{\theta}(A_t | S_t)$ can be interpreted as the direction in the parameter space that increases the probability of taking action A_t in state S_t . The policy gradient, then, is a weighted average of all possible directions to increase the probability of taking any action in any state, but weighted by reward signals, so that if taking a certain action in a certain state is associated with high reward, then that direction would be highly reinforced, and vice versa.

>> Section 4

If $b_i(S_t)$ is chosen well, such that $b_i(S_t) \approx \sum_{\tau=t}^T (\gamma \tau - t R_{\tau}) = \gamma V^{\pi_{\theta_i}}(S_t) - b(S_t)$, this could significantly decrease variance in the gradient estimation. That is, the baseline should be as close to the value function $V^{\pi_{\theta_i}}(S_t)$ as possible, approaching the ideal of: $\nabla J(\theta) = \mathbb{E}_{\pi_{\theta}} [\sum_{t=0}^T \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) (\sum_{\tau=t}^T (\gamma \tau R_{\tau}) - \gamma V^{\pi_{\theta}}(S_t)) | S_0 = s_0]$. Note that, as the policy π_{θ_t} updates, the value function $V^{\pi_{\theta_t}}(S_t)$ updates as well, so the baseline should also be updated. One common approach is to train a separate function that estimates the value function, and use that as the baseline. This is one of the actor-critic methods, where the policy function is the actor and the value function is the critic. The Q-function Q^{π} can also be used as the critic, since $\nabla J(\theta) = \mathbb{E}_{\pi_{\theta}} [\sum_{0 \leq t \leq T} \gamma^t \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) (Q^{\pi_{\theta}}(S_t) - V^{\pi_{\theta}}(S_t))]$.

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$$\nabla_{\theta} J(\theta) = E_{\pi_{\theta}} \left[\sum_{0 \leq t \leq T} \gamma^t \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \cdot Q^{\pi_{\theta}}(S_t, A_t) \mid S_0 = s_0 \right]$$

$$\nabla_{\theta} J(\theta) = E_{\pi_{\theta}} \left[\sum_{0 \leq t \leq T} \gamma^t \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \cdot (A_t - V^{\pi_{\theta}}(S_t)) \mid S_0 = s_0 \right]$$

by a similar argument using the tower law. Subtracting the value function as a baseline, we find that the advantage function $A^{\pi}(S, A) = Q^{\pi}(S, A) - V^{\pi}(S)$ can be used as the critic as well: $\nabla_{\theta} J(\theta) = E_{\pi_{\theta}} \left[\sum_{0 \leq t \leq T} \gamma^t \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \cdot A^{\pi_{\theta}}(S_t, A_t) \mid S_0 = s_0 \right]$

In summary, there are many unbiased estimators for $\nabla_{\theta} J(\theta)$, all in the form of: $\nabla_{\theta} J(\theta) = E_{\pi_{\theta}} \left[\sum_{0 \leq t \leq T} \gamma^t \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \cdot \Psi_t \mid S_0 = s_0 \right]$ where Ψ_t is any linear sum of the following terms: